

First Aid for Wounds and Severe Bleeding

This guide covers how to treat cuts, deep wounds, and severe bleeding when professional medical help is not immediately available. Based on guidelines from Mayo Clinic, Red Cross, and ANZCOR (Australian and New Zealand Committee on Resuscitation).

CALL FOR EMERGENCY HELP IMMEDIATELY for any wound that is: gaping open, deep, will not stop bleeding after 10 minutes of direct pressure, has an object embedded in it, or is on the head, chest, or abdomen.

SECTION 1: UNDERSTANDING TYPES OF BLEEDING

Venous bleeding (from veins): Dark red blood that flows steadily. Serious but generally easier to control with pressure.

Arterial bleeding (from arteries): Bright red blood that spurts or pulses with each heartbeat. This is the most dangerous type and can cause death within minutes. Requires immediate aggressive action.

Capillary bleeding (from small vessels): Slow oozing from minor cuts and scrapes. Usually stops on its own with minimal pressure.

CRITICAL FACT: A person can bleed to death from a large arterial wound in less than 5 minutes. Emergency services take an average of 7 to 10 minutes to arrive. Your immediate action is the difference between life and death.

SECTION 2: TREATING A WOUND — STEP BY STEP

Before you touch the wound:

Step 1. Check that you and the injured person are safe from ongoing danger. Do not approach if there is fire, live electrical wires, or structural collapse risk.

Step 2. Protect yourself. Put on gloves if available. If no gloves, use a plastic bag, clean plastic wrap, or several layers of clean cloth as a barrier between your hands and the blood. This protects both you and the injured person from infection.

Step 3. Call for emergency help or send someone to call. Tell them the location, the nature of the injury, and how much blood has been lost.

Stopping the bleeding:

Step 4. Do NOT remove clothing or debris from a deep wound unless it is very obviously on the surface. Removing embedded objects can cause catastrophic bleeding — the object may be plugging the wound.

Step 5. Cover the wound with the cleanest material you have — sterile gauze, a clean cloth, a clean shirt, or a clean towel. The priority is to stop bleeding, even if the material is not perfectly sterile.

Step 6. Apply firm, direct pressure using the palm of your hand. Press hard and continuously. Do not press on an eye injury or an embedded object.

Step 7. Hold the pressure for a FULL 5 to 10 minutes without lifting the cloth to check. Every time you lift the cloth, you disturb the clot that is trying to form and restart the bleeding clock.

Step 8. If blood soaks through the cloth, do NOT remove it. Add more cloth on top and press harder. Removing the first cloth tears the forming clot.

Step 9. If possible, raise the injured limb above the level of the heart. Gravity reduces blood flow to the area and helps slow bleeding.

Step 10. If bleeding slows or stops, wrap the wound firmly (not so tight as to cut off circulation) with a bandage or strips of cloth to maintain pressure while you get help.

DO NOT: Apply powder, ointments, or any cream to a deep wound. Do not probe the wound. Do not attempt to clean a deep wound — this is a hospital task. Do not give the injured person food or water if surgery may be needed.

SECTION 3: WHEN DIRECT PRESSURE IS NOT ENOUGH — TOURNIQUETS

A tourniquet is used only for life-threatening bleeding from an arm or leg that cannot be controlled by direct pressure, or for traumatic amputation.

How to apply an improvised tourniquet:

Step 1. Use a tourniquet only on arms or legs. It cannot be used on the head, neck, chest, or abdomen.

Step 2. Place the tourniquet 5 to 7 centimetres (2 to 3 inches) ABOVE the wound — between the wound and the heart. Never place it directly over a joint (elbow or knee).

Step 3. If you have no commercial tourniquet, use a belt, scarf, or strip of cloth at least 4 centimetres wide. Narrow materials like rope or wire can cause tissue damage.

Step 4. Wrap the material around the limb twice. Tie a half-knot (like the first half of tying shoelaces).

Step 5. Place a stick, pen, or rigid object on top of the knot. Tie a second half-knot over the stick to hold it in place.

Step 6. Twist the stick until the bleeding completely stops. Expect the person to feel significant pain — this is normal and necessary.

Step 7. Secure the stick so it cannot unwind. Tie it in place with another strip of cloth.

Step 8. Write the exact time the tourniquet was applied on the person's skin or on a piece of tape attached to it. This is critical information for medical professionals.

Step 9. Do NOT remove or loosen the tourniquet once it has been applied. Only trained medical personnel should remove it.

TOURNIQUET WARNING: A tourniquet stops all blood flow to the limb. This is painful and causes tissue damage beyond 2 hours. It is a life-over-limb decision — use it when the alternative is death from blood loss.

SECTION 4: EMBEDDED OBJECTS IN A WOUND

If there is a knife, glass, nail, or other object embedded in the wound:

DO NOT remove it. The object may be plugging the wound and preventing catastrophic bleeding. Removing it may cause rapid, uncontrollable blood loss.

Step 1. Apply pressure around the object, not on it.

Step 2. Build padding (folded cloth) on each side of the object to stabilise it and prevent movement.

Step 3. Bandage carefully over the padding, keeping the object stable and protected.

Step 4. Get the person to medical care as quickly as possible.

SECTION 5: SIGNS OF SHOCK — RECOGNISE AND RESPOND

Severe blood loss causes shock, where the body cannot circulate enough blood to vital organs. This can be fatal. Watch for:

- Skin that is pale, cold, and clammy (sweaty)
- Rapid, weak pulse
- Fast, shallow breathing
- Confusion, dizziness, or loss of consciousness
- Extreme thirst
- Nausea or vomiting

What to do for shock:

Step 1. Lay the person flat on their back on the ground.

Step 2. Raise their legs about 30 centimetres (12 inches) above the level of their heart, unless there is an injury to the legs or spine.

Step 3. Keep the person warm with a blanket or clothing.

Step 4. Do NOT give the person food or water.

Step 5. Keep them still and calm. Reassure them that help is coming.

Step 6. Continue to monitor breathing. If breathing stops, begin CPR.

SECTION 6: WOUND INFECTION — SIGNS TO WATCH FOR

Even after bleeding is controlled, wounds can become infected. Seek medical help if any of the following develop:

- Increasing redness or warmth spreading outward from the wound
- Swelling that gets worse after 24 hours
- Pus (yellow, green, or foul-smelling discharge)
- Fever (feeling hot, shivering uncontrollably)
- Red streaks spreading away from the wound in a line — this indicates infection spreading through the lymph system and is a medical emergency
- The wound area becomes increasingly painful rather than improving